

Archaeological Remains of Food

The ways that archaeologists study food are structured by the limitations of the types of data we have available to us in the present. Archaeologists can only observe those material remains that survive in the archaeological record, for hundreds or thousands of years. So, what types of artifacts and ecofacts do archaeologists find and use to reconstruct diet in the past? And what methods do archaeologists use to study them? In future modules, we will explore in depth how these data types and methods allow archaeologists to ask and answer specific questions about food and society in the past.

Stone Tools

Stone tools are the earliest artifacts we find in the archaeological record, dating to more than 1 million years ago in Africa. These modified rocks were used for many purposes and become more specialized and complex over time. **Grinding stones** appear more recently in the archaeological record, and their use expanded considerably alongside agriculture, as they allowed grain to be ground into flour for various breads and porridges. Recent advances in the analysis of **microbotanical remains** allow archaeologists to recover **phytoliths** and **starch granules** from these objects, which, in turn, are used in the study of diet and food preparation. We will learn more about microbotanical remains, including phytoliths and starch granules, in a future module.



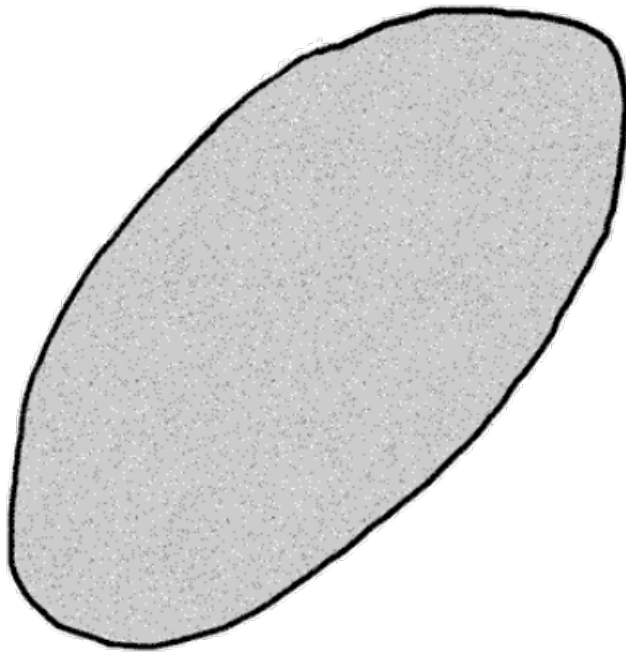
An Oldowan tool of the Lower Paleolithic (c. 1 million to 300,000 years ago), one of the earliest types of human-made tools. Found in Fonsorbes, southern France.

Source: [Wikimedia Commons](#)

(https://commons.wikimedia.org/wiki/File:Galet_am%C3%A9nag%C3%A9_MHNT.PRE.2011.0.483.jpg)

Start Gif

Stop Gif



Straight Flint

Double Bladed

Curved Flint

Curved Obsidian

Curved Oblique

Straight Serrated



Experimentally produced modern replicas of ancient sickles, used to harvest grains by cutting the stalk of the plant. Note the small, thin, sharp stone microblades mounted in the wooden handle.

Source: [Ethnoarchaeology](#)

(<https://www.tandfonline.com/doi/full/10.1179/1944289015Z.0000>)



Common Mesoamerican grinding stones: the mano (the handheld stone that is moved) and metate (the large stone slab that remains stationary). The grain or other item to be processed is placed between the stones. From Retalhuleu, Guatemala.

Source: [Wikimedia Commons](#)

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Grinding stones are natural stones shaped to produce rough, but flat or consistently curved, surfaces that can be used to grind seeds, fruits, or other food items, as well as minerals for other uses.

Microbotanical remains are the remains of plants that are so small as to require a microscope for identification and analysis. Examples include pollen, phytoliths, and starch granules.

Phytoliths are microscopic structures made of silica that plants deposit within and between plant cells. Phytoliths often have distinctive, three-dimensional shapes that allow experienced researchers to identify the plants in which they were produced.

Starch granules are structures created by plants to store starch for use as a future energy source. Starch granules are particularly common within storage organs, such as tubers and bulbs, and in seeds.

Animal Bones and Shells

Animal bones and shells are common features of archaeological sites from the earliest period of human evolution. Bones and shells are durable over long periods of time, and careful study of animal remains—the field known as **zooarchaeology**—allows archaeologists to identify the type of animal, as well as the specific part of the animal, from which a bone came. Zooarchaeologists can study large numbers of bones of domesticated animals from a specific site or a number of sites in an area to answer questions about how people managed animal herds for food and other products. We will learn much more about zooarchaeology in a future module!



A shell midden from coastal Argentina.

Source: [Wikimedia Commons](#)

(https://commons.wikimedia.org/wiki/File:Conchero_al_sur_de_Puerto_Deseado.jpg)





Cow upper leg bones (left pair: left, rear limb; right, front limb) and sheep upper leg bones (right pair: left, rear limb; right, front limb); note the differences in size and shape.

Source: [Zoomorphology](https://link.springer.com/article/10.1007/s00435-021-00516-6)

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Animal bones being sorted during initial stages of analysis. Like bones are placed together (e.g., all ribs are placed in a row in the front left)

Source: Dr. Marston

Zooarchaeology is the field of research that focuses on animal remains from archaeological contexts.

Pottery

One of the most durable types of artifact that humans have ever produced is pottery. Although ceramics were used for many purposes, the most common was the production of pots, bowls, and plates for the preparation and consumption of food, as well as large containers for food transport and storage. Pottery is a common facet of nearly every agricultural society ever discovered, and since pottery is both extremely durable (it rarely breaks down, even after thousands of years, since it is both dense and inorganic) and fragile (people accidentally break dishes all the time), it often dominates the contents of trash deposits we find archaeologically. By systematically studying the shapes and decoration of pottery, as well as its chemical and mineralogical composition, archaeologists are able to determine the likely uses of different types of pots, where they were produced, and how they moved during their lifespans. **Chemical-residue analysis**, a method we will study in detail in a future module, allows archaeological chemists to determine what compounds are present within a cooking or storage vessel and, thus, what foods or beverages were stored or cooked in those pots.



A Bronze Age beer jug; the strainer in the spout filters out grains and other residues from the fermented beer when one is drinking. From Tel Eton, Israel; in the Israeli National Museum.

Source: Dr. Marston



Aztec molcajete (grater bowl) used to prepare spices, sauces, and condiments; from the site of Calixtlahuaca, Mexico.

Source: [Arizona State University](https://calixtlahuaca.blogspot.com/2007/06/salsa-stone-and-pottery.html).

(<https://calixtlahuaca.blogspot.com/2007/06/salsa-stone-and-pottery.html>)



A pithos—a large storage jar—filled with its original contents: dried figs that were accidentally burned. From Ashkelon, Israel; in the Israeli National Museum.

Source: Dr. Marston



Pottery fragments at the Roman site of Karanis, Egypt.

Source: Dr. Marston

Chemical residue analysis is the field of research that focuses on the chemical traces of human activity in the past. Most residue analysis is conducted on artifacts, especially porous pottery that might have absorbed the chemicals of the items cooked or stored within the pot.

Plant Remains: Macroscopic and Microscopic

Plant remains that survive archaeologically come in several forms, but the study of any of these is known as **paleoethnobotany** (or sometimes archaeobotany). Most basically, we can divide plant remains into two categories: macroscopic, which we can see directly, and microscopic, which can only be seen with a microscope. Macroscopic remains include seeds and other large fragments of plants, which are often preserved because they have been incompletely burned, which has rendered them inorganic but still morphologically intact. Microscopic remains include structures that form within or between plant cells, such as phytoliths and starch granules, and plant sex cells—namely, pollen. We will learn much more about all of these types of plant remains when we study paleoethnobotany in detail in a future module.



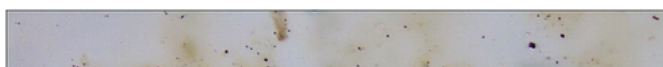
Carbonized (charred) seeds, mainly grape seeds, from a paleoethnobotanical sample taken from the site of Ashkelon in southern Israel.

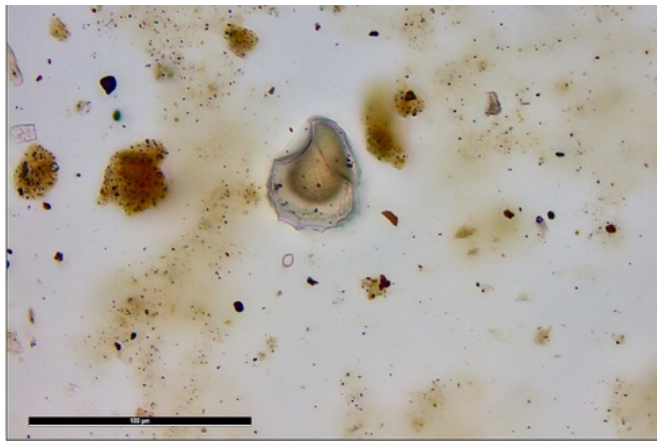
Source: Dr. Marston



Desiccated dates and date pits from the Roman city of Karanis, in the western desert of Egypt.

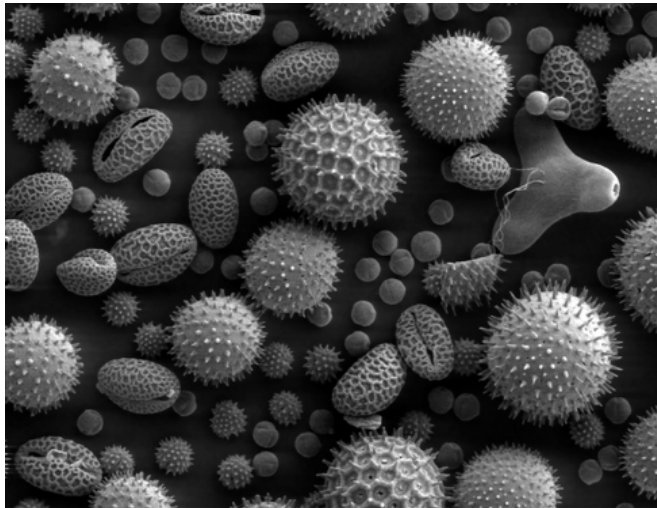
Source: Dr. Marston





A phytolith produced only within a specific group of grasses. From Bronze Age levels at the site of Megiddo, Israel.

Source: Dr. Marston



Pollen comes in an amazing variety of shapes and textures when seen, as here, under scanning electron microscopy. Magnification around 500x.

Source: [Wikimedia Commons](https://commons.wikimedia.org/wiki/File:Misc_pollen.jpg)

(https://commons.wikimedia.org/wiki/File:Misc_pollen.jpg)

Paleoethnobotany (also termed "archaeobotany") is the field of research that focuses on plant remains in archaeological contexts.

Human Remains

Human bones (and other tissues, when preserved) record much about our diets. The chemical and isotopic composition of the foods we eat structures the composition of our bodies, as well. **Stable-isotope analysis**—the study of the stable isotope ratios of carbon, nitrogen, and oxygen in an organic substance—allows archaeologists to determine several aspects of diet, including the degree of meat and seafood consumption; the presence or absence of specific plants, such as maize (corn), in the diet; and the locations in which plants and animals consumed were raised. Teeth are of particular value, as fossilized dental plaque—known as **dental calculus**—on our teeth preserves chemical and genetic remnants of the food an individual ate. **Chemical-residue analysis** on human tissues, such as hair, can reveal the consumption of certain drinks and drugs with unique chemical compositions, and the approximate time before death that those were consumed. We'll learn much more about these tools in a methods unit of a future module and in case studies we will explore later in the class.

The following contains images of human skeletal and mummified remains from archaeological excavations. It is not necessary to view these images to complete this course. To view these images, please click the button below:

☒ **A. Correct:** Yes, I would like to view these images.

Correct answer

☐ **B.** No, I would **not** like to view these images.



An adult human skeleton buried in an Early Iron Age tumulus in Albania.
Source: Dr. Marston



Dental calculus on archaeologically recovered human teeth.

Source: Nature Microbiology (<https://www.nature.com/nmicrobiol/>)



Naturally preserved mummy from the hyperarid North Coast of Peru.

Source: Wikimedia Commons

(https://commons.wikimedia.org/wiki/File:Peruvian_mummified_male,_c.1200-1400_Wellcome_L0035650.jpg)

Stable isotope analysis determines the relative presence of different isotopes within a substance. In archaeology, this is used for several purposes; in the archaeological study of human remains, it can allow us to determine what types of foods people ate in the past and where

they lived during specific periods of their growth and development.

Dental calculus is mineralized dental plaque that forms during life, especially in the absence of tooth brushing, and persists on teeth in archaeological contexts.